

Lecture 35: Bayes estimator - examples, priors

Example Consider $X \sim \text{Bin}(n, \theta)$, $\Theta = [0, 1]$, $\mathcal{X} = \{0, 1, \dots, n\}$; To estimate θ (unknown)
 Additionally, for Bayes estimator, assume, say,
 squared error loss function $L(\theta, d) = (\theta - d)^2$ and the prior distribution
 for θ , $\pi(\theta) \equiv \text{Beta}(\alpha, \beta)$, i.e., $\theta \sim \text{Beta}(\alpha, \beta)$, $\alpha, \beta > 0$, both known.

$\therefore$ Loss is squared error, Bayes estimator of θ is mean of the posterior distribution $\pi(\theta|x)$.

The j.t. dist_n of X and θ , $f(x, \theta) = f(x|\theta) \pi(\theta)$

$$= \binom{n}{x} \theta^x (1-\theta)^{n-x} \frac{1}{\text{Beta}(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$\therefore$ Marginal dist_n of X , $f_X(x) = \int_0^1 f(x, \theta) d\theta = \binom{n}{x} \frac{1}{\text{Beta}(\alpha, \beta)} \cdot \text{Beta}(\alpha+x, n-x+\beta)$

$\Rightarrow$ Posterior dist_n of $\theta|x$, $\pi(\theta|x) = \frac{1}{\text{Beta}(\alpha+x, n-x+\beta)} \theta^{\alpha+x-1} (1-\theta)^{n-x+\beta-1}$, $\theta \in [0, 1]$

$$\equiv \text{Beta}(\alpha+x, n-x+\beta)$$

$$\Rightarrow \hat{\theta}_B(x) = E(\theta|x) = \frac{\alpha+x}{n+\alpha+\beta}$$

$$\text{and } \hat{\theta}_{\text{Bayes}} = \frac{\alpha+x}{n+\alpha+\beta}$$

Note (a) Suppose it is a 'no prior' situation with $\alpha, \beta \rightarrow 0$, then

$$\hat{\theta}_{\text{Bayes}} = \hat{\theta}_{\text{UE}} = \frac{x}{n} \quad (\text{in fact, the UMVUE of } \theta)$$

(b) If it is a 'no data' situation, then

$$\min E(L(\theta, d)) = \min \int_0^1 L(\theta, d) \pi(\theta) d\theta$$

$\therefore$ Squared error loss fn, min at $\hat{\theta}_{\text{Bayes}} = \text{mean of } \pi(\theta)$

$$= \frac{\alpha}{\alpha+\beta} = \hat{\theta}_\pi, \text{ say}$$

$$\text{Observe, } \hat{\theta}_{\text{Bayes}} = \omega \hat{\theta}_\pi + (1-\omega) \hat{\theta}_{\text{UE}}, \text{ with } \omega = \frac{\alpha+\beta}{n+\alpha+\beta}$$

(Bayes estimator is a convex combination of 'no data' & 'no prior' estimators).

a numerical example.

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A coin is being checked whether it is unbiased or not.

Out of 5 independent tosses, 'head' appears 4 times.

Frequentists' approach Let $X = \#$ heads in 5 indep tosses.

Then $X \sim \text{Bin}(5, \theta)$, with $\theta = \text{prob. of head in a toss}$.

$$\text{and } \hat{\theta}_{\text{VE}} (\text{or } \hat{\theta}_{\text{UMVUE}}, \hat{\theta}_{\text{MLE}}) = \frac{X}{n} = \frac{4}{5} \quad (\text{the estimator})$$

with obsd $X=4$, the estimate is $\hat{\theta} = 0.8$.

Bayesian approach

In addition to $f(x|\theta) \equiv \text{Bin}(5, \theta)$, we ~~need~~ assume $\pi(\theta) \equiv \text{Beta}(1, 1)$ (prior belief is coin is unbiased $\therefore \hat{\theta}_\pi = \frac{1}{1+1} = 0.5$)

But, with the observed data, viz, $X=4$, this is revised to the posterior mean value and

$$\hat{\theta}_{\text{Bayes estimate}} = \frac{4+1}{5+1+1} = 0.714$$

(under squareserror loss) 'no data', or, 'prior only' estimate of 0.5

An upward correction to the 'no data', or, 'prior only' estimate of 0.5 because of large no. of heads.

Choice of priors

Conjugate prior: Let $X \sim f(x|\theta)$ and $\pi(\theta)$ be the prior distribution of θ on Θ . Then $\pi(\theta)$ is said to be a conjugate prior if the corresponding posterior distribution $\pi(\theta|x)$ belongs to the same family of distributions as $\pi(\theta)$.

e.g. ① $X|\theta \sim \text{Bin}(n, \theta)$, $0 \leq \theta \leq 1$; $\theta \sim \text{Beta}(\alpha, \beta)$, $\alpha, \beta > 0$
 $\theta|x \sim \text{Beta}(\alpha+x, n+\beta-x)$. $\therefore \text{Beta}(\alpha, \beta)$ is a conjugate prior for θ when $X \sim \text{Bin}(n, \theta)$.

② $X|\theta \sim N(\theta, \sigma^2)$, $\theta \in \mathbb{R}$; σ^2 known; $\theta \sim N(\mu, \tau^2)$, $\mu \in \mathbb{R}$, $\tau^2 > 0$ both ~~are~~ known,

then it is a conjugate prior with $\theta|x \sim N(a, b)$, where

$$a = \frac{\frac{n}{\sigma^2} \bar{x} + \frac{1}{\tau^2} \mu}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}, \quad b = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

③ $X|\theta \sim \text{Poisson}(\theta)$, $\theta > 0$; $\theta \sim \text{Gamma}(\alpha, \beta)$, $\alpha, \beta > 0$, known,
 a conjugate prior $\therefore \theta|x \sim \text{Gamma}(\alpha+x, \frac{\beta}{n+1})$.

Non-informative prior

$\pi(\theta)$ is said to be a non-informative prior if it contains no information about θ , i.e., the distribution does not favour any value of θ over other values & treats all parameter values equally.

Ex $\pi(\theta) \equiv U(0, a)$, $a > 0$, a uniform prior is a non-informative prior.
For $\theta \in (0, a)$, it is a proper prior.

A prior, in addition to being non-informative, can be improper, when the prior distribution is not a proper pdf/pdf.

Typically, π is some non-neg. weight fn, like,

$\int d\pi(\theta) = \infty$, e.g., when $\Theta = \mathbb{R}$, and $\pi(\theta) = 1$.

Jeffrey's prior. The prior density π is proportional to $\sqrt{|I(\theta)|}$ (just, $\sqrt{I(\theta)}$, in case of a single unknown parameter).

Ex. $X \sim \text{Bin}(n, \theta)$, then $I(\theta) = \frac{n}{\theta(1-\theta)}$ and

$\pi^*(\theta) \propto \{\sqrt{\theta(1-\theta)}\}^{-1}$, with normalizing constant

$\pi(\theta) \equiv \text{Beta}(\frac{1}{2}, \frac{1}{2})$, which is a proper prior in this case.